Het Patel

574-336-4929 | hetptl143324@gmail.com | linkedin.com/in/hetptldev | github.com/hetptl1433 | Portfolio

Summary

MS in Computer Science student with hands-on experience in software engineering, machine learning workflows, and AI output evaluation. Built full-stack applications and Python-based ML pipelines for experimentation, validation, and data-driven problem solving across research and product environments. Experienced in dataset preparation, model output review, debugging, result analysis, and documenting repeatable workflows, with interests in AI evaluation, data quality, and reliable AI-assisted systems.

Education

Illinois Institute of Technology — Chicago, IL
M.S. in Computer Science — Expected Dec 2026
Relevant Coursework: Machine Learning, Algorithms, Network Programming, Cryptography & Security

Sardar Vallabhbhai Institute of Technology — Anand, India
B.S. in Computer Science — May 2024

Experience

Research Assistant, Illinois Institute of Technology — Jan 2026 – Present

- Built and evaluated **Python/PyTorch machine learning pipelines** across **5+ prototype variants**, comparing output quality, runtime behavior, and performance metrics to support model-selection decisions.
- Automated **dataset preparation**, experiment execution, and result aggregation with Python scripts, improving reproducibility and reducing repeated manual validation across multiple runs.
- Reviewed **model outputs**, identified preprocessing and configuration issues, and documented recurring error patterns to improve experimentation, debugging, and result quality.
- Organized experiment notes, evaluation results, and testing observations in a structured format for easier comparison, handoff, and repeatable analysis.

Teaching Assistant, Illinois Institute of Technology — Sep 2025 – Jan 2026

- Supported 40+ students through labs, grading, and office hours, applying consistent evaluation criteria and giving clear, structured feedback on technical work.
- Explained machine learning and software engineering concepts step by step, helping students improve correctness, clarity, and problem-solving approach.

Software Engineer / Machine Learning Engineer, Barodaweb — Jun 2024 – Dec 2024

- Contributed to a **camera-integrated digital signage platform** that used **computer vision** and image-processing logic to estimate viewer attributes and trigger personalized content in near real time.
- Built and tested **full-stack web application components** using **React, Node.js, Express, and MongoDB** for data capture, device-linked workflows, configuration handling, and operational dashboards across multiple deployments.
- Reviewed system behavior under different lighting, positioning, and site conditions, documented failure patterns, and helped validate fixes before release.
- Tuned thresholds, dwell-time rules, and trigger logic through iterative testing, implementing **10+ reliability improvements** across the workflow.
- Worked with cross-functional teammates to document setup steps, configuration updates, and test outcomes for smoother deployment and support handoffs.

Software Engineer Intern, VMC — Mar 2024 – May 2024

- Used C# and .NET to analyze workflow defects, validate fixes, and document rollout steps for an online ticketing platform.
- Reproduced user-facing issues, reviewed system behavior, and communicated findings clearly for implementation and QA follow-up.

Projects

CryptexLLM — Python, PyTorch

- Built an **LLM-powered time-series forecasting pipeline** for BTC using OHLCV market data, patch-based tokenization with normalization (RevIN), and a frozen LLM reprogramming head to predict price moves.
- Optimized for trading relevance by training with **direction-aware losses (MADL/GMADL)**, then validated with **walk-forward optimization** to report out-of-sample accuracy, trend behavior, and risk-adjusted returns under realistic fees.

Credit Card Fraud Detection Using Transformer — Python, PyTorch, scikit-learn

- Built a **Transformer-based credit card fraud detector** (FT-Transformer, PyTorch) on 284,807 transactions with 0.17% fraud prevalence; achieved **ROC-AUC 0.9643**, **Recall 0.8667**, and **Accuracy 0.9929**.
- Improved performance with a **Feature-Gated FT-Transformer** (Conv1D + sigmoid token gating), boosting **ROC-AUC to 0.9827** and reducing test loss from **0.3360 to 0.2758**.

Real-Estate Management — React, Mapbox GL, GraphQL

- Built a Real-Estate Parcel Manager using React + Mapbox GL to visualize property parcels on an interactive map, enabling search/filter workflows and smooth pan/zoom UX for spatial decision-making.
- Implemented a GraphQL data layer to query parcel attributes such as parcel ID, zoning, and area fields, improving scalability and maintainability versus ad-hoc REST calls.

E-Motel — React, Node.js, Express, MongoDB, Socket.IO, AWS, Vercel

- Built **E-Motel**, a **full-stack motel management web app** using the MERN stack (MongoDB Atlas, Express, React, Node.js) with Tailwind CSS, supporting room and booking tracking, customer records, and staff operations.
- Implemented **real-time updates with Socket.IO** and deployed a production build on **Vercel/AWS**, improving operational visibility and supporting reliable workflow coordination.

Skills

Languages: Python, C/C++, JavaScript, TypeScript, SQL, Java, C#
ML & Data: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, transformers, model evaluation, dataset preparation, data preprocessing, experiment design, error analysis, structured testing
AI Tools: ChatGPT, OpenAI Codex, Claude, Google Gemini, prompt engineering, AI-assisted debugging, AI-assisted research
Software & Web: React, Node.js, Express.js, MongoDB, REST APIs, GraphQL, Socket.IO, HTML/CSS, Tailwind CSS
Tools & Workflow: Git, Docker, AWS, Vercel, Linux, VS Code, Postman, debugging, technical documentation, automation scripting
Professional: Attention to detail, analytical thinking, communication, collaboration, organization, reliable execution